Date: April 10, 2025

A33HF

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240 Uran Street
Hillsdale, MI 49242

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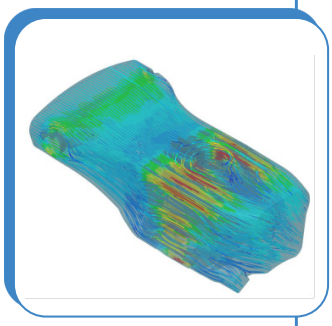
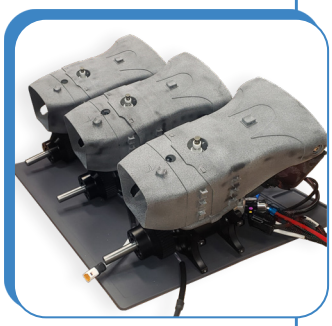
www.cobra-aero.com





DESCRIPTION

The A33HF Propulsion System from Cobra AERO is the same core engine as the A33N with special modifications to optimize heavy fuel operation.

- High power to weight
- Low acoustic signature
- Additively manufactured cylinder, mount, intake, and exhaust system unlock design potential that other manufacturing methods can't (performance, weight, package, sound, efficiency)
- Hardware, electronics and software all designed specifically for UAS applications
- Machined crankcase with integrated starter-generator
- Integrated focal isolation mount
- Active cooling duct for optimal operating temperatures across a variety of ambient conditions
- Parallel hybrid capabilities boost output when bursts of climb or sprint power are required
- Combustion system was designed specifically for Heavy Fuel - NOT a HF conversion.

SPECIFICATIONS

Engine Type:	Air-cooled 2-stroke single
Displacement:	33 cc (2.0 ci)
Mass:	3.15 kg (6.94 lbs) (Includes core engine, exhaust, focal isolation mount, ECU, fuel pump and accumulator, injector, ignition, generator, harnesses)
Power Output:	1.9 kW (2.5 HP) at 8050 RPM
PreHeat:	10 minutes @ 110 Watts
Fuel Consumption:	500 g/kW-hr (0.83 lb/hp-hr) cruise
Generator Output:	200W (400 W intermittent)
Fuel Type:	JP5, JP8, Jet A, VP heavy fuel 50:1 premix

SYSTEM INCLUDES

- Power4Flight IntelliJect ECU
- Self-priming fuel pump and accumulator
- All required EFI sensors and wire harnessing
  - *CHT, MAT and Fuel pressure*
  - *Baro and MAP*
  - *CDI module*
- Integrated preheating system
- Electric power generator
- Cooling duct with active control
- Optional IntelliGen PMU for starting, charging, and power augmentation capabilities
- Available NDAA-848 compliant ECU

FEATURES

- Altitude compensation
- Onboard data logging, maintenance schedule, and logbook
- Full size spark plug for high altitude performance and improved MTBO
- Plug n' Play with multiple autopilots
- Free communications ICD and software developers kit for CAN and serial
- Free user interface for engine and configuration development

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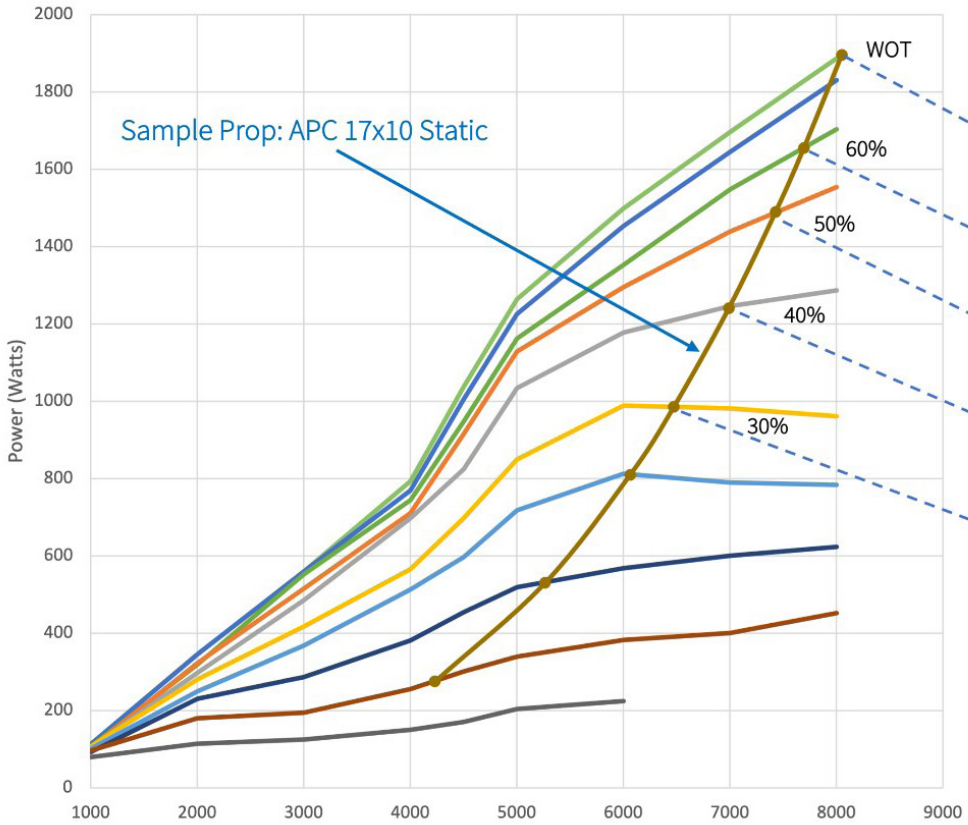
NOTE: All performance data is taken with an AC dynometer. Fuel flow is measured with a precision positive displacement fuel flow meter, and A/F ratio is set using a five gas emissions bench.

A33HF Performance
Preliminary



Power kW (hp)	BSFC g/kW-hr (lb/hp-hr)
1.89 (2.53)	733 (1.20)
1.65 (2.21)	625 (1.03)
1.49 (2.00)	617 (1.01)
1.24 (1.66)	564 (0.93)
0.98 (1.31)	502 (0.83)